

Figure 9: Biplot

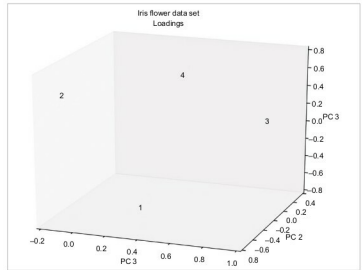


Figure 11: Loadings 3D

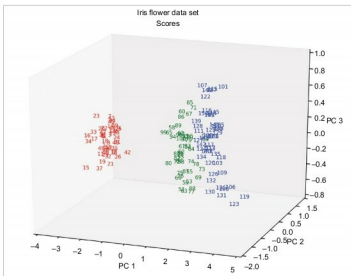


Figure 10: Scores 3D

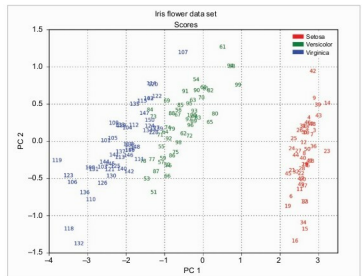


Figure 12: Scores SVD

Principal component analysis

This second example is about three different techniques—matrix algebra, singular value decomposition (SVD) and modular toolkit for data processing (MDP). About the first, the covariance matrix is calculated on the scaled data. Then, Eigenvalues and Eigenvectors are calculated from the covariance matrix. Lastly, the Eigenvalues and Eigenvectors are sorted. The scores are calculated by a dot product with the data scaled and the Eigenvectors. The percentage of variance is explained and its running total is also calculated.

```
covmat=cov(dataS, rowvar=False)
eigval,eigvec=linalg.eig(covmat)
idx=eigval.argsort()[::-1]
eigval=eigval[idx]
eigvec=eigvec[:,idx]
scores=dot(dataS,eigvec)
percentage=[0]*w.ncols
runtot=[0]*w.ncols
```

```
for i in range(0,w.ncols):
    percentage[i]=eigval[i]/sum(eigval)*100
runtot=cumsum(percentage)
```

The number of components (N), the variance explained by each component (VAR), its percentage (PCT) and the percentage running total (SUM) can be presented as a table. This table can be drawn using the package *prettytable*. The results are formatted with a certain number of decimal figures and then each column is added to the table.

```
from prettytable import *
o=range(1,w.rows+1)
v=range(1,w.ncols+1)
e=["%.2f" % i for i in eigval]
p=["%.4f" % i for i in percentage]
r=["%.2f" % i for i in runtot]
pt=PrettyTable()
pt.add_column("N",v)
pt.add_column("VAR",e)
```